

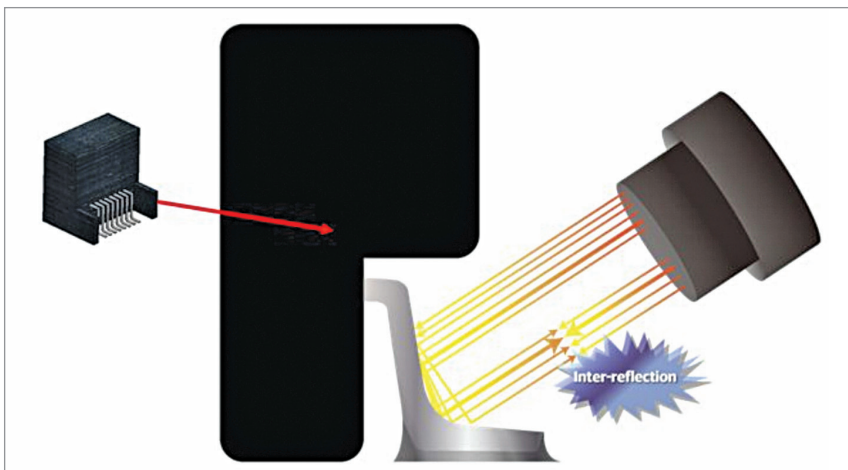


Fig. 3: Single projection's problem with specular areas (Source: Koh Young Technology)

“To avoid faulty solder joints, defects, false calls, and costly rework, we may need easier 3D solder paste inspection as well as 3D AOI inspection,” says Olivier Pirou, Head, Product Management and Marketing at Mycronic, an AOI manufacturing company.

Most of the AOI solutions still offer a limited level of 3D coverage for components and solder joints. But a full 3D coverage is needed to identify defects. These full-coverage 3D AOI systems not only inspect common defects, such as missing or wrong components, but accurately provide true volumetric height, identify coplanarity, and lead-bridging issues.

Some 3D AOI solutions have used the multi-frequency shadow moiré technology. Unlike 3D AOI systems using a brightness threshold, the moiré technology uses the height threshold to extract critical body and the lead tip information. This data allows the AOI to provide reliable body and joint measurements with CAD dimensions. It can locate the body properly in the first stage of the inspection process.

However, 3D inspection systems also have disadvantages. This approach cannot check for printed part nomenclature, polarity marks, or colour. Even after using multiple digital projectors, shadowing may occur. Therefore, some additional program-

ming may be required to train inspection components. Using 3D inspection can increase cycle times as projectors may need more time to project multiple fringe patterns needed for image processing. And lastly, machine costs are comparatively high as the technique is not yet mature.

Shadow problems

The shadow problem is inevitable even in 3D inspection machines. When various components are assembled on a board, some short parts like resistors may be shadowed by tall neighbouring components like connectors. This may lead to inaccurate 3D measurement results.

Many manufacturers use an 8-way projection mechanism shown in Fig. 1 that could possibly eliminate blind spots and construct a true 3D image by using multiple projectors.

Board warpage

Sometimes, on a warped board, pad positions defined by board CAD or Gerber files appear distorted. This may confuse conventional inspection systems and generate incorrect data. The board warp becomes a prominent issue during the reflow process because of high temperatures.

Creating new advanced algorithms can possibly address this challenge.

Inter-reflection and solder joint problems

Inter-reflection is a common issue in most optical systems, where light itself becomes the source of information. In the PCBs, soldered and tinned components on assembled boards may have specular surfaces, which reflect some lighting back to the camera while creating strong inter-reflection with other lighting reflections. In such a case, some reflected light does not reach the camera and generates false signals that provide incorrect measurement details.

Some AOI systems today overcome this with the use of superior multi-view sensors. Using these multi-view sensors and advanced 3D image fusion algorithms, it is possible to obtain error-free measurements.

Measuring a solder joint is a complicated task as each solder deposit may vary according to the design and printing conditions. Specular reflections in these will not allow to detect solder defects.

Regular challenges in AOI

- Greater need for programming, which is complicated and cumbersome
- Capacitance values are different and the specifications and colours are the same, which can easily lead to missed judgment
- Different character processing methods cause large differences in polarity judgment accuracy
- Most of the AOIs' understanding of virtual welding is ambiguous, resulting in missed judgment
- There is a problem in detection of Pingyang, shielding cover, and shielding point
- It is impossible to detect solder joints where BGA, FC, and other flip-chip components are not visible
- The detection speed of most AOI products is slow. Speed of a few AOI using scanning methods is faster, but the rate of misjudgment is higher